

Financial Literacy for Students

Porter Clevidence

Antonio Duran-Ramos

Parth Gajjar

Arpit Vora

May 6, 2025

List of Contents

List of Figures	1
List of Tables	2
Project Plan	3
Introduction	3
Team Roles and Responsibilities	3
Methods and Techniques	3
Data	3
Data Description and Analysis	4
Dataset Statistics	4
Text Processing	5
Tokenizer Definition	5
Surface Level Normalization Definition	5
Stopping Definition	5
Stemming/Lemmatizing Definition	5
Index Construction	6
Term Weighting Definition	6
Index Design	6
Retrieval Model	7
Ranking	7
Implementation	8
Source Code	8
Video Demonstration	8
Discussion	9
References	10

List of Figures

List of Tables

Project Plan

Introduction

This project aims to build a search engine that allows students to retrieve relevant documents from a financial dataset. It is part of an educational initiative to promote financial literacy. This enables students to better understand and handle real-world financial situations encountered during their professional lives. To achieve this goal we have designed and implemented an *Information Retrieval (IR)* system that processes textual data, builds an *index* over a document collection, and ranks documents in response to user queries.

Team Roles and Responsibilities

Porter Clevidence

- Implemented BIM search
- Implemented BM25 search
- Implemented Language Model retrieval
- Implemented VSM search
- Search algorithm selection

Antonio Duran-Ramos

- Development environment setup
- Graphical interface design
- LaTeX report setup
-

Parth Gajjar

Arpit Vora

Methods and Techniques

In our system design we chose to use a combination of **Docker** and **uv**, along with various Python libraries. Docker allowed us to standardize our development environment so that our search engine behaved identical, regardless of host machine. **uv** allowed us to

formalize our project as it handles initializing Python virtual environments and it manages dependencies for reproducible Python environments. As for the Python libraries we utilized, **pandas** was used to read from the provided JSON files to create structured data frames. **scikit-learn** was used for english stop words and its functions: **TfidfVectorizer** and **cosine_similarity**. **rank-bm25** was used for its implementation of the BM25 ranking algorithm. **streamlit** was used to implement the graphical interface for our search engine.

Data

The dataset provided consisted of a collection of financial documents, along with static queries and their corresponding relevance judgments.

Data Description and Analysis

Dataset Statistics

Text Processing

Tokenizer Definition

Surface Level Normalization Definition

Stopping Definition

Stemming/Lemmatizing Definition

Index Construction

Term Weighting Definition

Index Design

Retrieval Model

Ranking

Implementation

Source Code

Video Demonstration

Discussion

References

[1]

[2]

[3]